

PROJECT AND TEAM INFORMATION

Project Title

Smart Driver Drowsiness & Safety System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Driver drowsiness and distraction are important road safety concerns, especially during long journeys, night driving, and monotonous driving conditions. A driver may become tired or lose attention for a short period without realizing the risk, reducing reaction time and increasing the possibility of accidents. The Smart Driver Drowsiness & Safety System is proposed as an integrated driver-assistance solution. It will monitor the driver's condition, detect signs of drowsiness and distraction, generate progressive alerts, control music when required, maintain event history, provide a safety score, and send an emergency message to a registered family contact during critical situations.

The project is motivated by the need for a practical and affordable safety system that can improve driver awareness. It also provides a real-world application for integrating Data Structures, C++, Operating System concepts, and DBMS in a single safety-oriented application.

State of the Art / Current solution (1 pt)

Modern vehicles and driver-assistance systems use cameras and sensors to monitor driver behaviour and identify possible fatigue or loss of attention. Some systems provide visual or audible warnings when signs of drowsiness are detected. However, basic or low-cost solutions often focus mainly on one function, such as eye-closure detection or an alarm.

The proposed system combines multiple safety functions in one prototype. It will monitor drowsiness and distraction, provide progressive alerts, control music during unsafe conditions, maintain event history, calculate a simple safety status, and support an emergency family message. Data structures will manage safety events and alerts, while a database will store driver, contact, and event information.

Project Goals and Milestones (2 pts)

The main goal is to develop a smart driver safety prototype that detects drowsiness and distraction and responds according to the severity of the detected condition.

Major goals: 1. Detect prolonged eye closure and identify possible drowsiness. 2. Detect prolonged driver distraction. 3. Implement progressive warning and critical alerts. 4. Automatically reduce or pause music during unsafe conditions. 5. Send an emergency message to a registered family contact during critical situations. 6. Store drowsiness, distraction, and alert events. 7. Generate a Good, Warning, or Critical safety status.

Milestones: 1. Requirement analysis, literature study, architecture, and module design. 2. Drowsiness and distraction detection module. 3. Alert management, music control, and emergency message module. 4. Queue, stack, linked-list and SQLite based data management. 5. Full integration, testing, debugging, and prototype preparation.

Project Approach (3 pts)

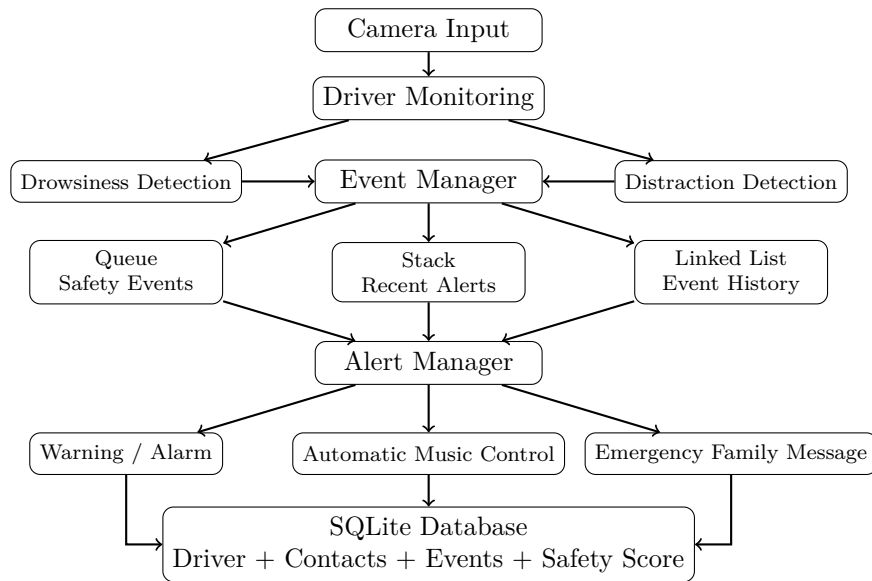
The system will be developed using a modular approach. A camera will provide input for monitoring the driver's face and eye activity. The monitoring module will analyse the driver's condition and generate safety events when drowsiness or distraction is detected.

A progressive alert mechanism will classify events by severity. Initial unsafe conditions will generate warnings, while repeated or prolonged conditions will generate stronger alarms. A critical condition will activate an emergency message to the registered family contact. Music will be reduced or paused during unsafe conditions so that alerts remain clearly audible.

A queue will handle incoming safety events, a stack will maintain recent alerts, and a linked list will maintain event history. C++ classes and objects will implement separate modules. Operating System concepts will be applied through concurrent processing for monitoring and alert handling. SQLite will store driver details, emergency contacts, event history, alerts, and safety records.

The prototype will use C++, OpenCV, SQLite, and C++ threads, with development and version control through VS Code and GitHub. Modules will be tested separately before final integration.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The expected outcome is a functional prototype of a Smart Driver Drowsiness & Safety System. The system will monitor the driver's condition, detect possible drowsiness and distraction, and generate progressive safety alerts. Music control will support clear alert delivery, while critical situations will provide an emergency message to the registered family contact.

The system will maintain safety event history and generate a simple Good, Warning, or Critical status. Deliverables will include source code, database design, system architecture, working prototype, test cases, documentation, and a GitHub repository.

Assumptions

The system assumes that a camera is available and positioned to clearly capture the driver's face. Adequate lighting is assumed for reliable detection. The driver is assumed to remain reasonably visible to the camera. Emergency contact information is assumed to be available in the database. Internet connectivity may be required for sending an external family notification.

References

1. OpenCV Documentation – Computer Vision Library. 2. SQLite Documentation – Lightweight Database Engine. 3. C++ Reference – Standard C++ Language and Library Documentation. 4. Operating System concepts related to threads and synchronization. 5. Research literature on driver drowsiness and distraction detection.